Deep Learning for Medical Image Classification

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Applied Machine Learning-Advanced

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1 Introduction and Project Objective

This project uses transfer learning in PyTorch to classify pediatric chest X-rays as either **NORMAL** or **PNEUMONIA**. Rather than just chasing a high test score, the main goal is to understand which design choices actually drive performance and whether the model learns meaningful medical features instead of relying on dataset shortcuts.

The study is organized into four main steps:

1. **Dataset audit:** An initial analysis shows that raw image geometry (such as aspect ratio and size) already leaks strong signals about the target class.

2. **Validation:** A patient-grouped validation scheme is set up to tune hyperparameters, pick the best checkpoint, and calibrate the decision threshold without data leakage.

3. **Benchmarking and ablations:** The model is tested against both classical and deep learning baselines, followed by targeted ablations to isolate the impact of architecture choice, class imbalance handling, padding strategies, and learning rates.

4. **Interpretability and error analysis:** Grad-CAM maps and error breakdowns examine where the model focuses its attention, with a specific look at how padding affects predictions across the full test set.

2 Architectural Selection

The backbone is **DenseNet121**. The main reason is its track record on chest radiographs. CheXNet, a 121-layer densely connected network, reported state-of-the-art results on all 14 thoracic disease categories of ChestX-ray14, pneumonia included. That result belongs to a different benchmark and a different population, so how well the model does on this pediatric dataset has to be established by the evaluation that follows.
Inside a dense block each layer receives the feature maps of all the layers before it. This **dense connectivity** favours feature reuse, helps information and gradients flow through the network and keeps the parameter count low. It fits chest radiographs well, since a decision can depend both on fine texture and contrast and on larger anatomical structures. **ImageNet-pretrained weights** give a reasonable starting point for a development pool of limited size, and the network is small enough to be fine-tuned with the available hardware. It is adapted to the binary task with a new classification head and progressive unfreezing (Section 4).

3 Dataset and Exploratory Data Analysis

3.1 Source, population, and prediction target

The data are the chest-radiograph part of the public dataset released by Kermany et al., made of X-rays of children aged one to five acquired during routine care at Guangzhou Women and Children's Medical Center, China. Each image is labelled **NORMAL** ($y = 0$) or **PNEUMONIA** ($y = 1$), with bacterial and viral cases merged into the second class. The subtype can be read from the filenames and is used only for description. Filenames, directory names and subtype strings never reach the model.

3.2 Archive inventory and split composition

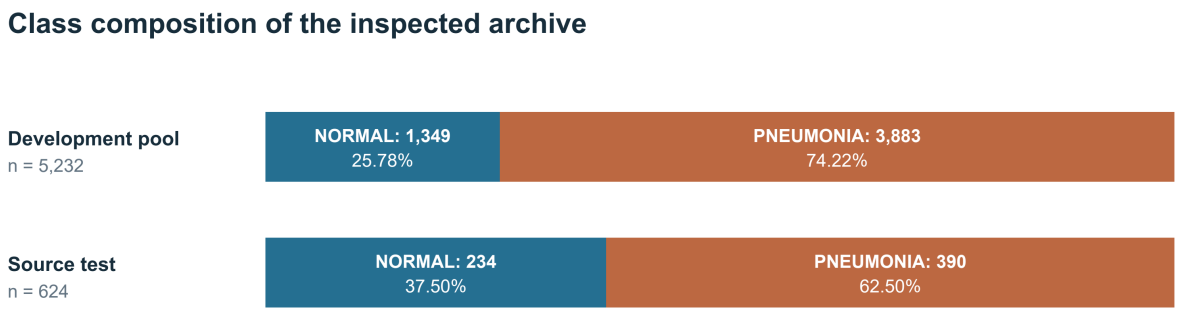
Counts were taken directly from `ChestXRay2017.zip`, the archive has a `train` directory and a `test` directory, but no validation set.

Table 1: Composition of the source archive. Bacterial and viral columns partition the PNEUMONIA class. Counts are images, not verified unique patients.

Source subset	NORMAL	Bacterial	Viral	Total	Pneumonia
Development pool	1,349	2,538	1,345	5,232	74.22%
Source test	234	242	148	624	62.50%
Entire archive	1,583	2,780	1,493	5,856	72.97%

Randomly splitting the dataset by image while keeping the class balance (`train_test_split` with `test_size=0.2` and `random_state=42`) gives **4,185** fitting images (1,079 NORMAL, 3,106 PNEUMONIA) and **1,047** validation images (270 NORMAL, 777 PNEUMONIA), with the 624 test images left untouched. This split ignores patients and is used only as a point

of comparison in experiment Q0 (Section 5). In every case augmentation is applied after splitting, and only to the fitting subset.
Pneumonia prevalence is **11.72 percentage points** higher in development than in test, which means an imbalance of about 2.88:1 against 1.67:1. A classifier that always answers PNEUMONIA would reach 74.22% accuracy in development and 62.50% in test, with zero specificity and 50% balanced accuracy. The same two prevalences, roughly 0.742 and 0.625, are also the precision expected from a random ranking. For this reason accuracy and precision-based metrics are always read together with class counts, sensitivity and specificity.



3.3 Image dimensions, intensity, and technical variability

Across the raw, unaugmented dataset, native image widths range from 384 to 2,916 pixels, heights from 127 to 2,713 pixels, and aspect ratios (width-to-height) from 0.835 to 3.379. The vast majority of scans (5,573 out of 5,856) are single-channel grayscale. The remaining 283 are stored as three-channel RGB, and all 283 belong to the training **PNEUMONIA** class. While this points to a potential confounder during image acquisition or export, it does not necessarily mean the images contain true color information or that a model would exploit it. To prevent unintended shortcuts, all images are converted to grayscale and replicated across three channels when required by the network backbone.

Table 2 breaks down these properties across splits and classes. We measure dimensions directly on the raw files. To make intensity and contrast comparable across images of drastically different resolutions, we first convert each scan to grayscale, resize it to 256×256 pixels, and normalize pixel values to $[0, 1]$. We define mean intensity as the average pixel value across the entire frame, and contrast as the standard deviation of those values. Because these metrics include borders, background canvas, and non-pulmonary anatomy, and are inherently affected by resampling, they should be interpreted purely as global image statistics rather than direct measures of lung tissue density.

Table 2: Image properties by source subset and class. Width and height are medians with interquartile ranges, the other columns are medians. N = NORMAL, P = PNEUMONIA.

Subset/class	Width, px [IQR]	Height, px [IQR]	W/H	Mean	Contrast
Development N	1,638 [1,466–1,820]	1,325 [1,152–1,541]	1.223	0.479	0.241
Development P	1,168 [1,000–1,368]	776 [640–968]	1.488	0.483	0.214
Test N	1,762 [1,533–2,030]	1,318 [1,040–1,639]	1.353	0.487	0.238
Test P	1,111 [992–1,256]	736 [632–872]	1.506	0.469	0.215

The most striking distinction between classes comes down to basic geometry: **NORMAL** images tend to be larger and less elongated than **PNEUMONIA** scans. In the development set, both classes share a similar median intensity, but they diverge noticeably in contrast. None of these patterns reflect actual pathology; they are far more likely artifacts of different scanner settings, cropping choices, export pipelines, or patient positioning. Resizing images to a fixed input size does not eliminate this problem either, as interpolation artifacts, padding borders, and the overall field of view still preserve clues about the original dimensions.

To quantify this risk, we tested native image width as a trivial predictor, simply ranking narrower images as more likely to show pneumonia. Evaluated across all positive–negative pairs (splitting ties evenly), this heuristic yields a rank AUC of **0.8797** on the development set and **0.9500** on the test set—without training a network, tuning a threshold, or inspecting pixel content. This alone calls for caution when interpreting suspiciously high performance early in training. That said, it does not prove which features DenseNet relies on in practice, and because the test-set AUC was checked post hoc, it should be viewed as an exploratory diagnostic rather than an independent benchmark.

3.4 Representative images and qualitative inspection

Figure 1 shows two development images for each category (NORMAL, bacterial and viral PNEUMONIA), namely the ones closest to the 25th and 75th percentiles of mean intensity, with ties broken by path order. The rule was chosen to show technical variability without picking examples by model confidence or by how easy they look. Aspect ratios are preserved and no contrast enhancement is applied.

The images are useful for checking exposure, thoracic coverage, orientation, burned-in annotations and structures outside the lungs. They only carry the source label, however, and not an expert localisation of the lesions. Opacification can be a relevant sign, but overall brightness or low contrast is not pneumonia by itself, and a visible line or bright object should not be given a clinical meaning without an expert's review.

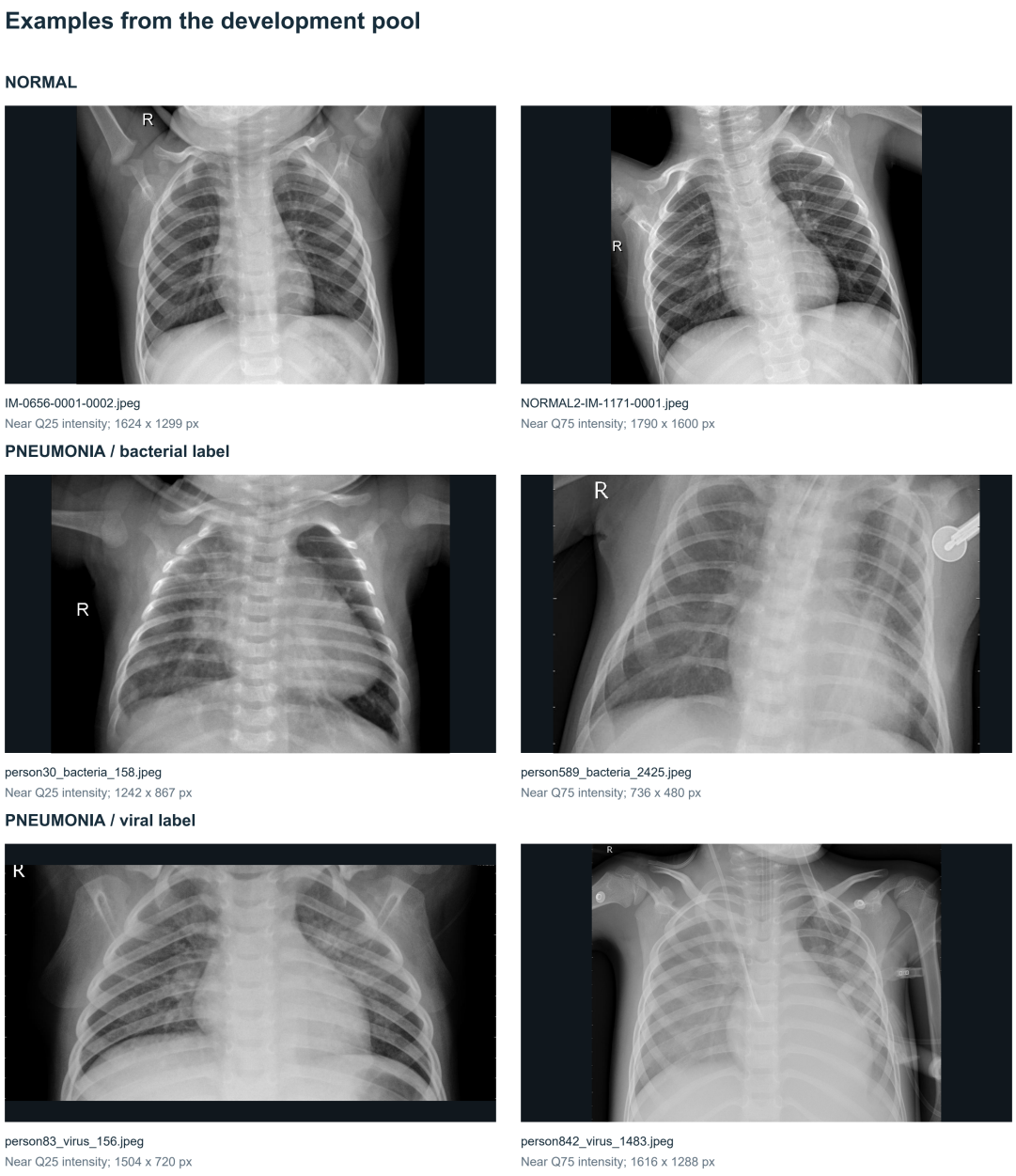


Figure 1: Six source images chosen with a fixed intensity-quantile rule. Rows are NORMAL, bacterial PNEUMONIA and viral PNEUMONIA, columns are examples near the lower and upper intensity quartiles.

3.5 Duplicate audit and limits of patient separation

SHA-256 hashing identified 30 duplicate groups totaling 62 images—meaning 32 redundant copies. Twenty-four groups were in the development set and six in the test set. None crossed between the two sets, and none had conflicting labels.

However, removing exact duplicates does not guarantee independent splits. A simple

random split can still place near-identical images or multiple scans of the same patient into both training and validation sets. To prevent data leakage, exact duplicates were removed and validation data was split strictly by patient identifier (Section 5).

Filenames provide an additional, cross-check, generic IDs like person123 overlap across bacterial and viral folders, with 170 appearing in both development and test sets. When including the disease subtype in the identifier, 4,097 unique groups (726 containing multiple images) were found, none of which crossed between sets. For normal images, prefixes like IM or NORMAL2-IM were preserved, because these patterns reflect file-naming conventions rather than verified patient records, they cannot prove patient identity.

4 Methods

4.1 Pipeline

Images are decoded in grayscale, replicated to three channels, resized so that the longest side matches the input size and then padded to a square. During training, random affine transformations, brightness and contrast perturbations and CLAHE are added, while validation and test share a single deterministic transform. DenseNet121 is initialised with ImageNet weights, and its classifier is replaced by dropout (0.4) followed by a linear layer that outputs one logit. The backbone stays frozen for the first three epochs and is then fine-tuned with AdamW, using three parameter groups (early backbone, late backbone, head). All settings are in `configs/default.yaml`. Table 3 lists the reference configuration together with the alternatives tested in Section 6.

Table 3: Reference configuration and the alternatives tested in screening.

Component	Reference	Alternatives (one at a time)
Backbone	DenseNet121, ImageNet	ResNet50, small CNN from scratch
Fine-tuning	3 frozen epochs, then all layers	linear probe, no frozen phase
Learning rates	3×10^{-6} / 7×10^{-6} / 4×10^{-4}	backbone $\times5$, head /4
Loss	focal, $\gamma = 2$, no class weight	BCE, BCE + `pos_weight`, focal + α_t
Sampling	$\propto 1/\sqrt{n_c}$	uniform, $\propto 1/n_c$
Padding	reflection	constant (mean), anisotropic resize
Validation split	patient-grouped	image-level (comparison only)

4.2 Class imbalance: an explicit choice

The development pool contains about 2.9 PNEUMONIA images for every NORMAL one. There are three common ways to handle this: re-sampling the batches, weighting the positive class in binary cross-entropy through a `pos_weight`, or adding a class weight to the focal loss. They are easy to stack by accident, and a focal loss with one α applied to both classes does not re-weight anything at all, it only rescales the loss. To evaluate these strategies clearly,

each approach is tested on its own (Q2):

$$\mathcal{L}^{w}_{\mathrm{BCE}} = -\tfrac{1}{B}\sum_i \big[w_+\, y_i \log p_i + (1-y_i)\log(1-p_i)\big], \qquad w_+ = n_0/n_1,$$
$$\mathcal{L}_{\mathrm{focal}} = \tfrac{1}{B}\sum_i \alpha_{t,i}\,(1-p_{t,i})^{\gamma}\,\ell_i, \qquad \alpha_{t,i} = \alpha y_i + (1-\alpha)(1-y_i),$$

where ℓ_i is the sample cross-entropy and $p_{t,i}$ is the predicted probability for the ground-truth class. Setting $\alpha = n_0/(n_0+n_1) \approx 0.26$ gives NORMAL cases roughly three times the weight of PNEUMONIA cases.
All four configurations get validation ROC-AUC values between 0.999 and 1.000 (Table 5), differences on par with run-to-run random variation. Because none of the alternatives improved ranking performance, we keep the baseline setup ($1/\sqrt{n_c}$ sampling and unweighted focal loss) and handle the trade-off between false positives and false negatives by tuning the decision threshold on validation data.

5 Experimental Protocol

Patient-grouped validation. Because the dataset lacks patient IDs, patient keys are derived directly from filename patterns, ensuring all images from the same individual are grouped together (including both bacterial and viral pneumonia cases). After removing exact duplicates, validation is performed on one fold of a 5-fold `StratifiedGroupKFold` split (Table 4). For comparison, the reference setup is also trained on a standard image-level split (`ref_image_split`) to measure how much data leakage inflates validation scores.

Table 4: Split composition after duplicate removal. No patient key is shared between training and validation.

Split	Images	Patient keys	NORMAL	PNEUMONIA	Pneumonia %
train	4,171	2,289	1,077	3,094	74.2
val	1,035	572	271	764	73.8
test	618	426	231	387	62.6

Selection rules fixed in advance. (i) For each run, the checkpoint is the epoch with the highest validation ROC-AUC. (ii) The final configuration is the one with the highest mean validation ROC-AUC in the screening. (iii) The decision threshold is the highest value that still gives a validation sensitivity of at least 0.95, since a missed pneumonia is treated as worse than a false alarm. The threshold is frozen before the final evaluation, and test results play no role in any of these choices.
Screening runs use 384×384 inputs, eight epochs and a single seed. The selected configuration and the reference are then retrained at 512×512 with three seeds and reported as mean $\pm$ standard deviation.

6 Experimental Comparison

6.1 Baselines

Three non-deep baselines share the exact same data split, thresholding rule, and evaluation metrics as the neural models: a dummy constant predictor (ROC-AUC 0.5 by construction), a logistic regression trained solely on image geometry (native width, height, and aspect ratio), and a logistic regression on 64×64 downsampled thumbnails. The geometry-only baseline achieves a validation ROC-AUC of 0.921 and a test ROC-AUC of 0.882. This, rather than a trivial 0.5, sets the real baseline any neural model must beat. In addition, three deep baselines—a small CNN trained from scratch, a DenseNet121 linear probe, and a standard ResNet50—are included in the initial screening.

6.2 Screening results

Table 5 and Figure 2 report validation results across all configurations, each modifying a single factor against the baseline. Test scores were also computed but played no role in decision-making, and are referenced only to clarify cases where validation performance reaches saturation.

Table 5: Screening on the patient-grouped validation set (384×384, 8 epochs). AUC is ROC-AUC and AP is average precision. Sensitivity, specificity and balanced accuracy are measured at the validation-selected threshold. The configuration in bold is the one selected for the final stage. The test set is not used here.

Question	Configuration	AUC	AP	Sens.	Spec.	Bal. acc.
Q0 reference	Original configuration, patient-grouped split	0.999	1.000	0.952	1.000	0.976
	Original configuration, image-level split	0.999	1.000	0.986	0.993	0.989
Q1 architecture	DenseNet121, no frozen phase	1.000	1.000	0.987	1.000	0.993
	ResNet50, same schedule as reference	0.998	0.999	0.953	1.000	0.976
	4-block CNN trained from scratch	0.991	0.997	0.950	0.974	0.962
	DenseNet121, frozen backbone for all epochs (linear probe)	0.990	0.996	0.950	0.945	0.947
Q2 imbalance	Unweighted BCE, inverse-frequency sampler	1.000	1.000	0.976	1.000	0.988
	Unweighted BCE, uniform sampling	1.000	1.000	0.970	1.000	0.985
	Focal loss with class weight alpha=0.26 on PNEUMONIA, uniform sampling	0.999	1.000	0.950	1.000	0.975
	BCE with pos_weight = n_neg/n_pos, uniform sampling	0.999	1.000	0.962	1.000	0.981
Q3 padding	Anisotropic resize to a square, no padding	0.999	1.000	0.965	1.000	0.982
	Constant (mean-intensity) padding instead of reflection	0.999	1.000	0.971	1.000	0.986
Q4 learning rate	Backbone learning rates x5	**1.000**	1.000	0.988	1.000	0.994
	Head learning rate /4	0.999	1.000	0.975	1.000	0.988

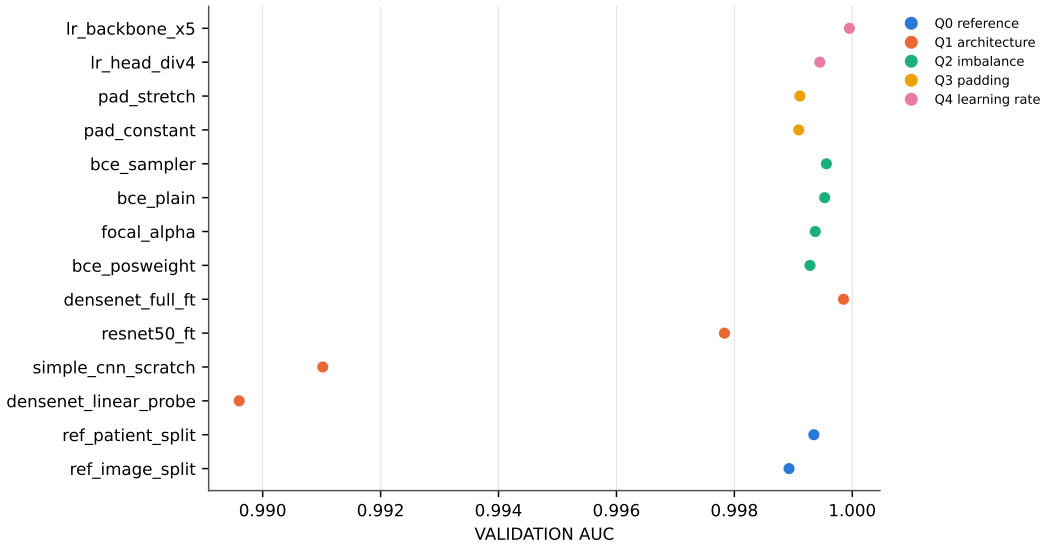


Figure 2: Validation ROC-AUC for each configuration, coloured by research question. Where present, error bars are standard deviations over seeds.

Q0 – leakage. The reference configuration reaches validation ROC-AUC 0.999 with the image-level split and 0.999 with the patient-grouped one. The two values are the same, so shared patients and exact duplicates are not what makes validation look almost perfect. Validation stays saturated even when patients are kept apart. The real drop happens between the development pool and the test directory (test ROC-AUC 0.965 for both runs), and the geometry baseline moves in the same direction (0.921 on validation, 0.882 on test). Any validation set drawn from the development pool is going to be optimistic, however it is grouped.

Q1 – architecture and transfer learning. Every model goes above validation ROC-AUC 0.99, so validation hardly separates them, although the ordering is consistent. Fine-tuned DenseNet121 (0.999–1.000) is ahead of the linear probe (0.990), which means frozen ImageNet features are not enough and the gain comes from adapting the backbone. The gaps are wider on the test set. The scratch CNN gets 0.906, no better than logistic regression on thumbnails (0.913), the linear probe gets 0.959 and fine-tuned DenseNet121 between 0.965 and 0.967. ResNet50 comes close (0.998 on validation, 0.955 on test) but is never ahead, which supports DenseNet121, if only by a small margin. Removing the frozen warm-up has no measurable effect.

Q2 – imbalance. ROC-AUC barely moves (0.999–1.000), but the threshold needed to reach validation sensitivity 0.95 goes from 0.44 with the inverse-frequency sampler to 0.81 with plain BCE. Once these thresholds are applied, all four variants give test sensitivity between 0.984 and 0.992 and specificity between 0.69 and 0.75, which is within single-seed noise. Re-weighting and re-sampling shift the score distribution, but here they do not improve discrimination.

Q3 – padding. Grad-CAM (Section 8) shows positive attribution inside the reflected borders, which is what motivated this test. Neither alternative changes validation ROC-AUC (0.999 in both cases). On the test set both are slightly above the reference, with 0.974 for constant padding and 0.973 for anisotropic resizing against 0.965, from a single seed each. Grad-CAM statistics were only computed for the final model, so the effect of these variants on attribution was not measured.

Q4 – learning rates. Multiplying the backbone learning rates by five gives the highest validation ROC-AUC (0.99995), so rule (ii) selects it. Dividing the head rate by four gives 0.9994. The difference is in the fourth decimal place, below the variability between seeds, so the rule is applied but the choice does not mean much. Within the tested range, performance is not sensitive to the learning rates.

7 Final Model: Test Evaluation

The selected configuration is *Backbone learning rates x5*. Over 3 seeds it reaches test ROC-AUC 0.957 ± 0.009. At the threshold chosen on validation ($t = 0.610$) the test sensitivity is 0.992 and the specificity 0.651.
For comparison, the reference configuration (Table 3), specifically the set-up of the first submission retrained under the present protocol, is evaluated in the same way. The selected model differs from it only in the backbone learning rates.

Model	AUC [95% CI]	AP	Sens.	Spec.	F1	ΔAUC vs geometry
Logistic regression on native width/height/aspect	0.882 [0.853, 0.908]	0.902	0.987	0.442	0.851	–
Constant score (training prevalence)	0.500 [0.500, 0.500]	0.626	1.000	0.000	0.770	-0.382 [-0.409, -0.353]
Logistic regression on 64x64 thumbnails	0.913 [0.890, 0.935]	0.946	0.977	0.524	0.864	+0.031 [+0.004, +0.059]
Backbone learning rates x5	0.957 ± 0.009 [0.938, 0.973]	0.966	0.992	0.651	0.903	+0.065 [+0.037, +0.095]
Original configuration, patient-grouped split	0.954 ± 0.002 [0.933, 0.971]	0.960	0.991	0.667	0.905	+0.074 [+0.046, +0.103]

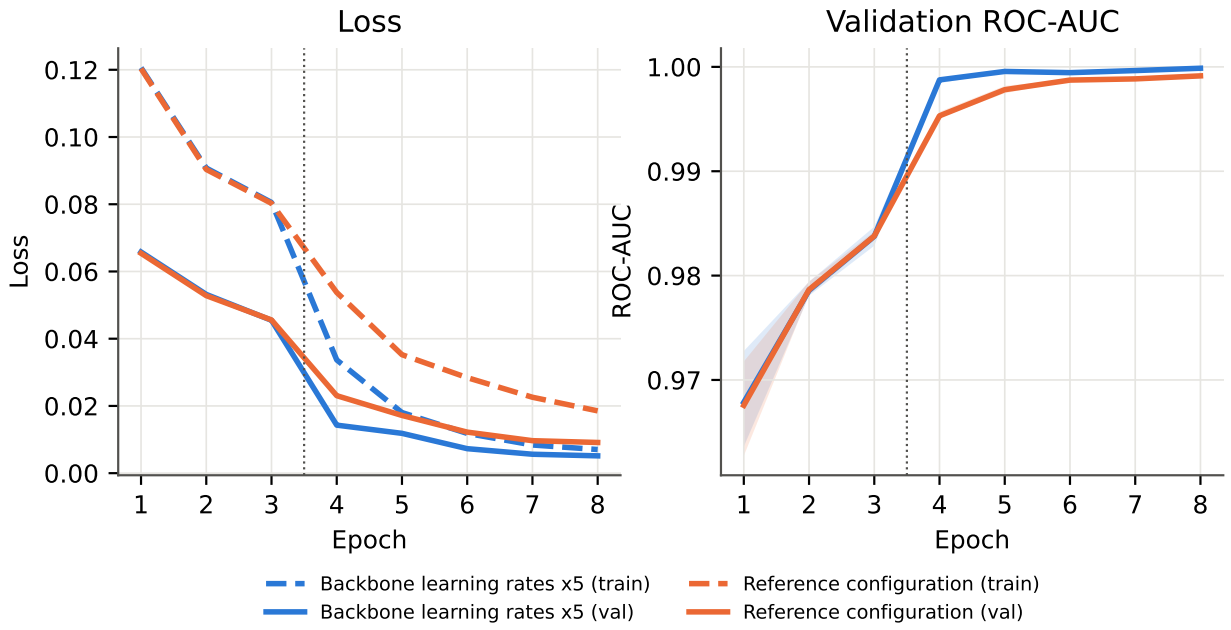


Figure 3: Learning curves of the final runs, averaged over seeds, with the band going from minimum to maximum. Dashed lines are the training loss on augmented, re-sampled data and solid lines are validation. The dotted vertical line marks the start of backbone fine-tuning.

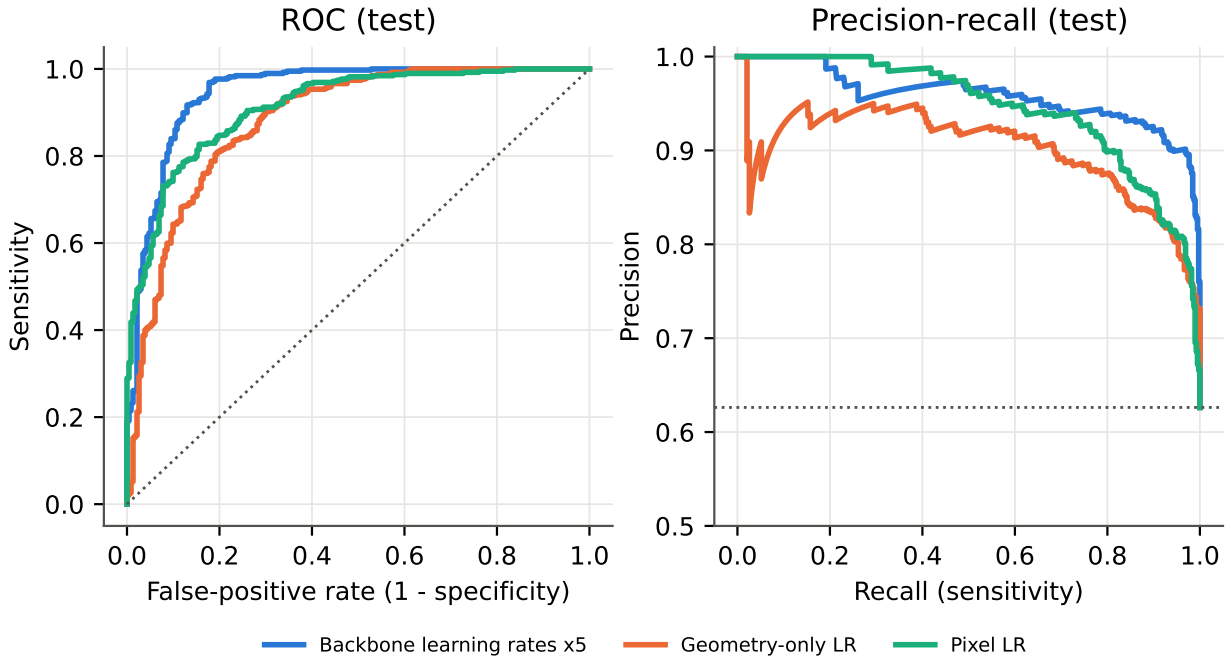


Figure 4: Test ROC and precision–recall curves of the final model (first seed) and of the geometry and pixel baselines. Dotted lines show chance (ROC) and prevalence (PR).

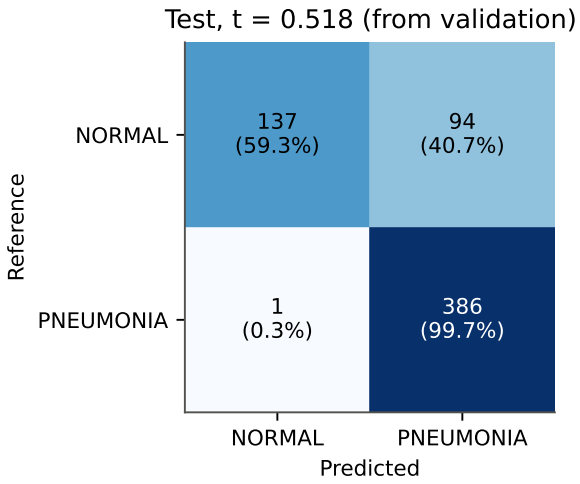


Figure 5: Test confusion matrix of the final model at the frozen threshold. Percentages are normalised by row, so the diagonal shows sensitivity and specificity.

It is tempting to move the threshold after looking at the test results. Raising it from 0.5 to 0.7, for example, trades a few missed pneumonias for fewer false alarms (Table 6). A threshold picked that way would be tuned on the test set, though, so the table is purely descriptive and the reported operating point stays the one selected on validation.

Table 6

Model	Threshold	Sensitivity	Specificity	FN	FP
Selected (backbone LR ×5)	0.5	0.996 ± 0.001	0.570 ± 0.039	1.7	99.3
Selected (backbone LR ×5)	0.6	0.992 ± 0.000	0.633 ± 0.035	3.0	84.7
Selected (backbone LR ×5)	0.7	0.989 ± 0.003	0.697 ± 0.040	4.3	70.0
Selected (backbone LR ×5)	0.8	0.985 ± 0.003	0.756 ± 0.027	5.7	56.3
Reference	0.5	0.992 ± 0.002	0.612 ± 0.029	3.0	89.7
Reference	0.6	0.991 ± 0.001	0.685 ± 0.029	3.7	72.7
Reference	0.7	0.982 ± 0.006	0.769 ± 0.044	7.0	53.3
Reference	0.8	0.960 ± 0.007	0.840 ± 0.027	15.7	37.0

Both final configurations beat the geometry baseline, and their paired ΔAUC intervals exclude zero. On the test set, however, the selected model is no better than the reference (0.957 ± 0.009 against 0.954 ± 0.002) and it is less stable. Across seeds its threshold ranges from 0.41 to 0.90 and test specificity from 0.55 to 0.81, because a validation set that is almost separable leaves the threshold poorly determined. Most errors are false positives. On average 81 of the 231 NORMAL images are flagged, while only 3 of the 387 pneumonias are missed. In screening terms few cases slip through, but roughly one healthy child in three would be sent for further review.

Why specificity is low. Two factors contribute mainly. (i) Padding: neutralising it removes 92 of the 94 false positives and raises test ROC-AUC from 0.947 to 0.972, although

it also turns 170 of 386 true positives into misses at the same cutoff (Section 8). (ii) At ROC-AUC 0.957, keeping sensitivity above 0.99 costs a false-positive rate of about one third. The specificity of 0.859 reported in the first submission was obtained by choosing the threshold (0.7) on the test set, which was a mistake on my part: it made the result too optimistic. The values here come from thresholds fixed without looking at the test set. At the common cutoff 0.5 the earlier run still scored higher (0.705 against 0.61), split, checkpoint rule and test interpolation changed together, so that gap cannot be attributed to one cause.

8 Error Analysis and Explainability

8.1 Purpose and class-specific attribution

Although native dimensions separate the classes, they do not explain which visual features drive the classifier's predictions. We use Grad-CAM to visualize spatial evidence from the final DenseNet convolutional layer (16×16 maps for a 512×512 input).
With a single binary output logit z, the PNEUMONIA score is simply z, while the NORMAL score is $-z$. Channel importance weights are obtained by spatially averaging the gradients of the target class score. We then take the weighted sum of channels and upsample it to the original image dimensions. The resulting map highlights areas providing positive evidence for the selected class. For all padding-control comparisons, the target class is held fixed to the initial unpadded prediction. We differentiate with respect to logits rather than sigmoid outputs to prevent vanishing gradients caused by probability saturation.

8.2 Case selection and test-wide measurement

For the figures, the two most confident test cases of each outcome (TN, TP, FP, FN) at the frozen threshold are selected, without looking at their maps. The padding statistics, instead, are computed on *all* test images. Let L_{ij} be the value of the Grad-CAM map at pixel (i, j) and $|M|$ the number of padding pixels. For a non-zero map and a non-empty padding region, the share of attribution falling in the padding and its enrichment relative to the padding area are

$$m_{\text{pad}} = \frac{\sum_{i,j} M_{ij} L_{ij}}{\sum_{i,j} L_{ij}}, \qquad e_{\text{pad}} = \frac{m_{\text{pad}}}{|M|/(512^2)}.$$

An enrichment above one means that the padding receives more positive attribution per pixel than a uniform map would give it. It is a descriptive measure and not a causal contribution.

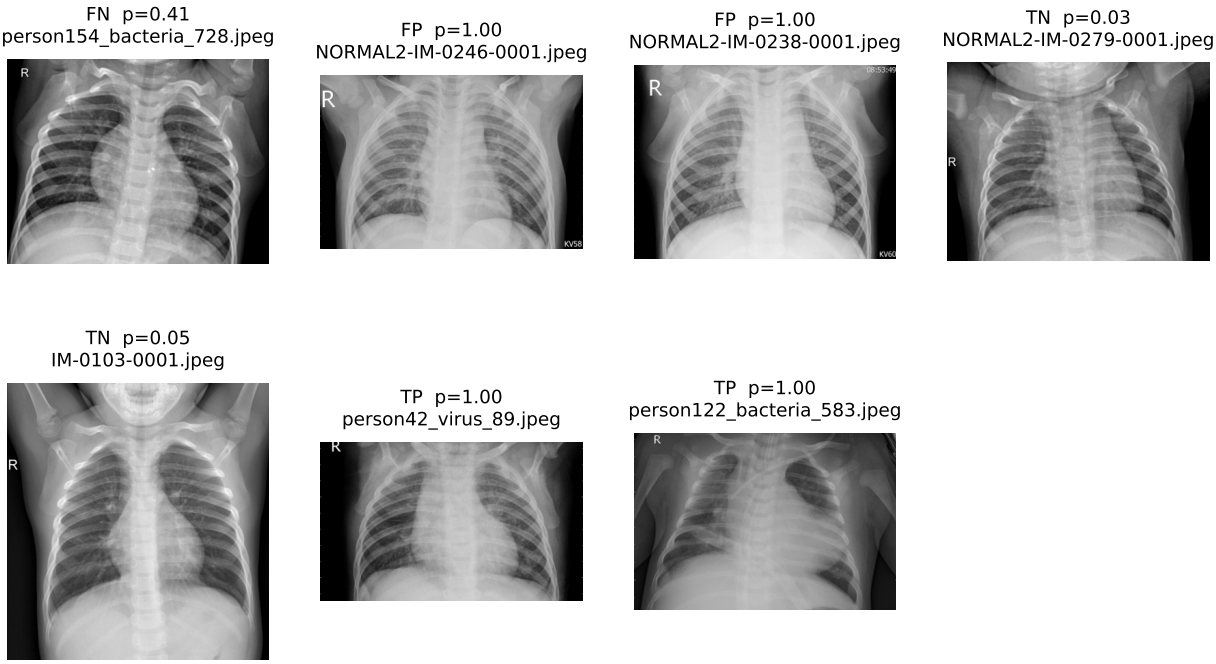


Figure 6: Most confident test cases of each outcome at the frozen threshold.

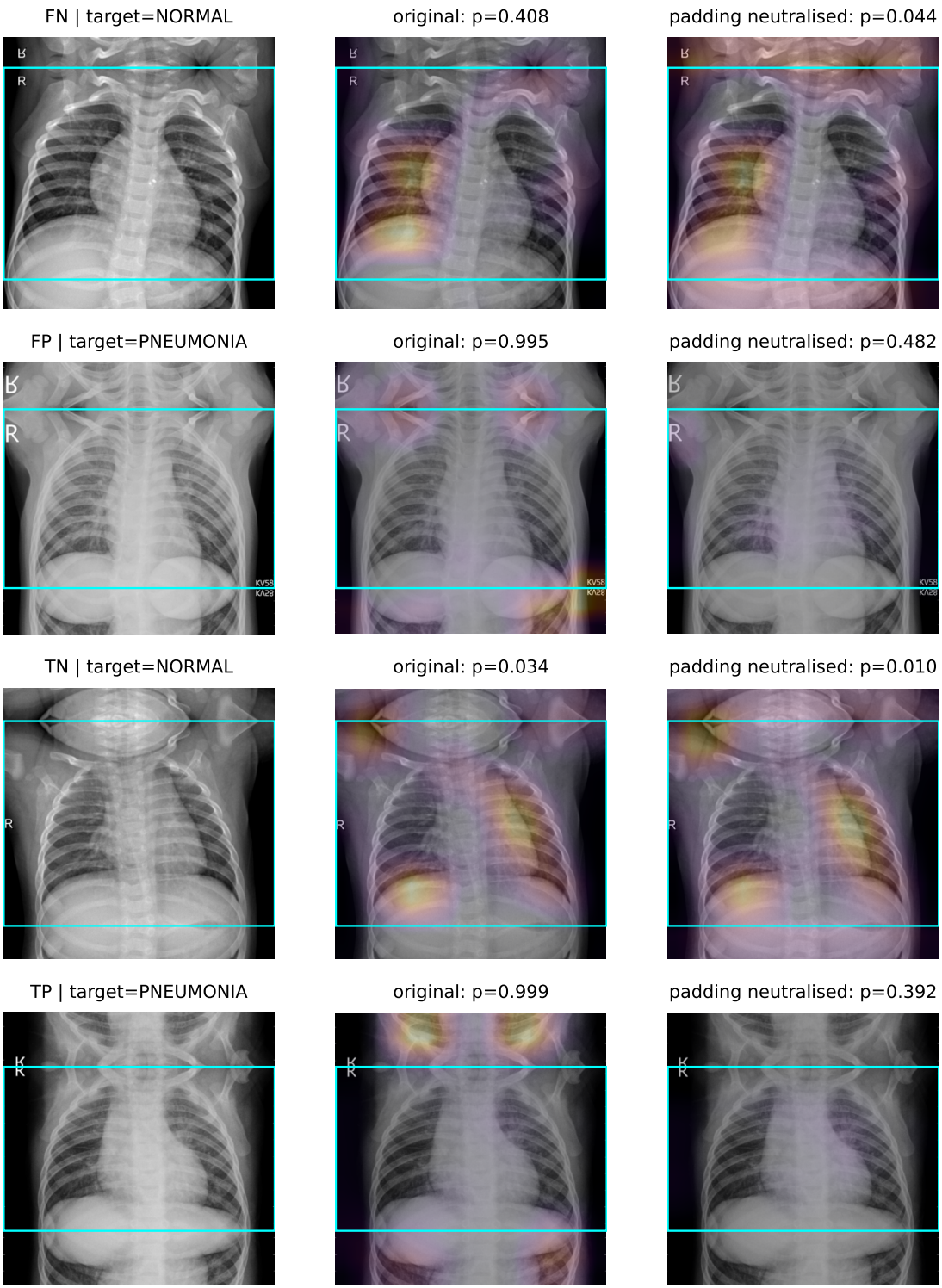


Figure 7: Grad-CAM for the predicted class on the original input and with the synthetic padding set to the mean intensity (same checkpoint, same target class). The cyan outline is the footprint of the resized image, not a lung contour. Each row uses one shared colour scale.

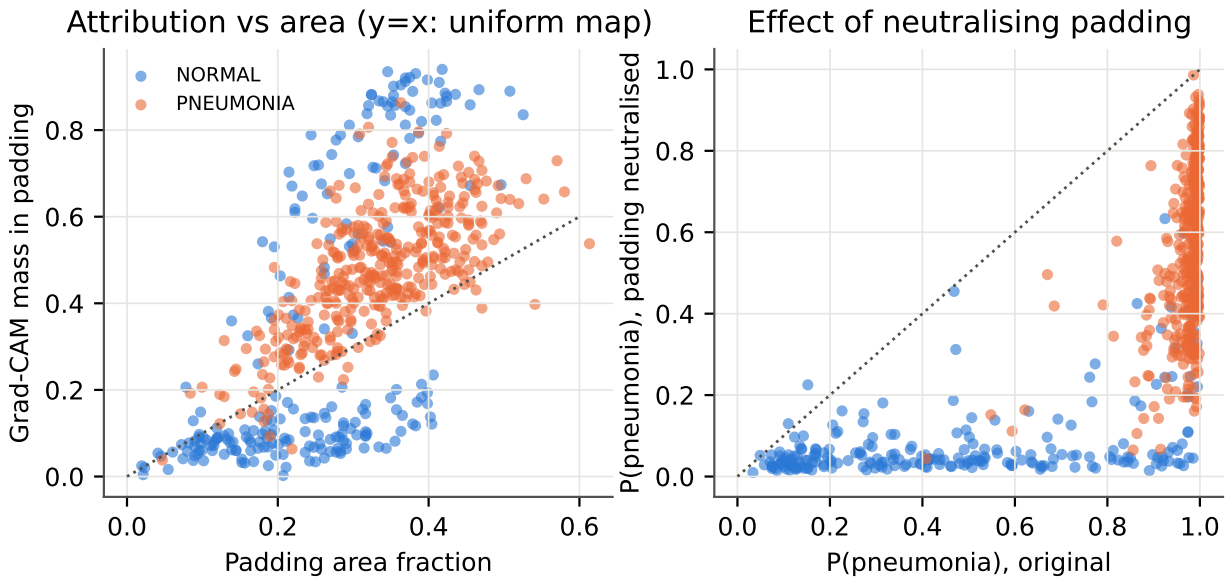


Figure 8: Padding analysis on all test images. On the left, share of Grad-CAM mass inside the padding against padding area (points above the diagonal concentrate attribution in the padding). On the right, PNEUMONIA probability before and after neutralising the padding.

In the final model (seed 42, reflection padding), synthetic padding accounts for a median 31% of the image area but absorbs 45% of the positive Grad-CAM attribution, showing enrichment (> 1) in 74% of test cases (median 1.38). The bias predominantly affects PNEUMONIA images (median padding mass of 0.48 versus 0.14 for NORMAL).

Neutralizing the padded regions reduces the average PNEUMONIA probability from 0.79 to 0.38, reclassifying 262 of 618 test cases (42%) exclusively from PNEUMONIA to NORMAL. This resolves 92 of 94 false positives at the cost of converting 170 of 386 true positives into false negatives. Strikingly, test ROC-AUC rises from 0.947 to 0.972 on neutralized inputs. Reflected borders thus inject an artificial upward shift into class scores without destroying the underlying anatomical signal, though the original decision threshold is rendered obsolete.

9 Limitations

- The data are pediatric and come from a single centre, and there is no external test set. The results do not carry over to adults or to other hospitals without new evidence.
- Validation is saturated (ROC-AUC ≥ 0.999 for most configurations), so model selection and the decision threshold are weakly determined. The threshold of the selected model ranges from 0.41 to 0.90 across seeds.
- Image geometry separates the classes. The geometry baseline and the padding analysis put bounds on the use of acquisition cues but do not rule it out.
- Neutralising the padding flips 42% of the final model's test predictions, so part of the decision depends on the reflected borders rather than on the radiograph.

- Grad-CAM works at 16×16 resolution and no lung masks are available, so attribution inside the image cannot be equated with pulmonary evidence.

10 Conclusion

This study highlights that native image dimensions alone yield a test ROC-AUC of 0.882, setting the non-trivial performance baseline that deep learning models must exceed. Under a patient-grouped split with decisions frozen on validation, the final setup achieves a test ROC-AUC of 0.957 ± 0.009, with sensitivity 0.992 and specificity 0.651.
Fine-tuning a pretrained DenseNet121 is clearly better than a linear probe or a CNN trained from scratch, and marginally ahead of ResNet50. Re-weighting strategies shift the operating point without improving ranking, which supports the unweighted focal loss with a threshold fixed on validation. The attribution analysis shows that reflected padding matters: neutralising it changes 42% of all test predictions (55% of the positive ones), cautioning against naive preprocessing.